

Then, $x + x - 5 = 45 \Rightarrow 2x = 50 \Rightarrow x = 25$.

So, number of girls = 25, number of boys = 20.

As calculated above, required average = 14.96 years.

Thus, I and III together also give the answer.

$\therefore$ Correct answer is (d).

32. I gives: $D + E = 14$... (i)

II gives: $A + B + C + F = 50 \times 4 = 200$... (ii)

III gives: $A + B + D + E = 40 \times 4 = 160$... (iii)

Adding (i) and (ii), we have: $A + B + C + D + E + F = 214$.

$\therefore$ Required average = $\left(\frac{214}{6}\right)$ years = 35.67 years.

Thus, only I and II together give the answer

$\therefore$ Correct answer is (a).

33. Let the three consecutive even numbers be x , $x + 2$ and $x + 4$.

I gives, $\frac{(x+4) + (x+6) + (x+8) + (x+10)}{4} = 17$

$\Rightarrow x + 7 = 17 \Rightarrow x = 10$.

So, the required numbers are 10, 12 and 14.

Thus, I alone gives the answer.

II gives, Difference between highest and lowest numbers = $(x + 4) - x = 4$.

This does not help to find the value of x .

So, II alone is insufficient.

III gives, $x^2 + (x + 2)^2 + (x + 4)^2 = 440$

$\Rightarrow x^2 + (x^2 + 4 + 4x) + (x^2 + 16 + 8x) = 440$

$\Rightarrow 3x^2 + 12x - 420 = 0$

$\Rightarrow x^2 + 4x - 140 = 0$

$\Rightarrow x^2 + 14x - 10x - 140 = 0$

$\Rightarrow (x + 14)(x - 10) = 0 \Rightarrow x = 10$.

So, the required numbers are 10, 12 and 14.

Thus, III alone also gives the answer.

$\therefore$ Correct answer is (d).

34. Let the four numbers in decreasing order be x , y , z and w . Then, we have to find whether the following statement is true or false:

$\frac{x+w}{2} > \frac{x+y+z+w}{4}$

I gives, $x - y < y - z$

II gives, $x - y > z - w$

$\Rightarrow x + w > y + z$

$\Rightarrow 2(x + w) > x + y + z + w$